

Al-Aqsa University
Faculty of Engineering
Dep. Intelligent Systems and
Computer Engineering



جامعة الأقصى
كلية الهندسة
قسم هندسة النظم الذكية والحاسوب

Pattern Recognition

Assignment 2: Wine Quality Classification Using SVM and XGBoost

Student name: Mohammed Al-masawabi

Student number: 1320222450

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1. Introduction

In pattern recognition, selecting an appropriate decision principle is critical for constructing robust classifiers. This report evaluates and compares two distinct machine learning paradigms for structural tabular data classification: **Support Vector Machines (SVM)**, which optimize geometric boundaries based on margin maximization, and **XGBoost**, an advanced ensemble method based on sequential decision trees.

To explore different boundary constraints on real-world physicochemical data, we implement two foundational SVM configurations:

1. **Hard Margin SVM:** A rigid formulation that tolerates no training errors ($C = 10^5$), assuming perfect linear separability.
2. **Soft Margin SVM:** A flexible formulation with a moderate penalty ($C = 1.0$) to handle overlapping data distributions.

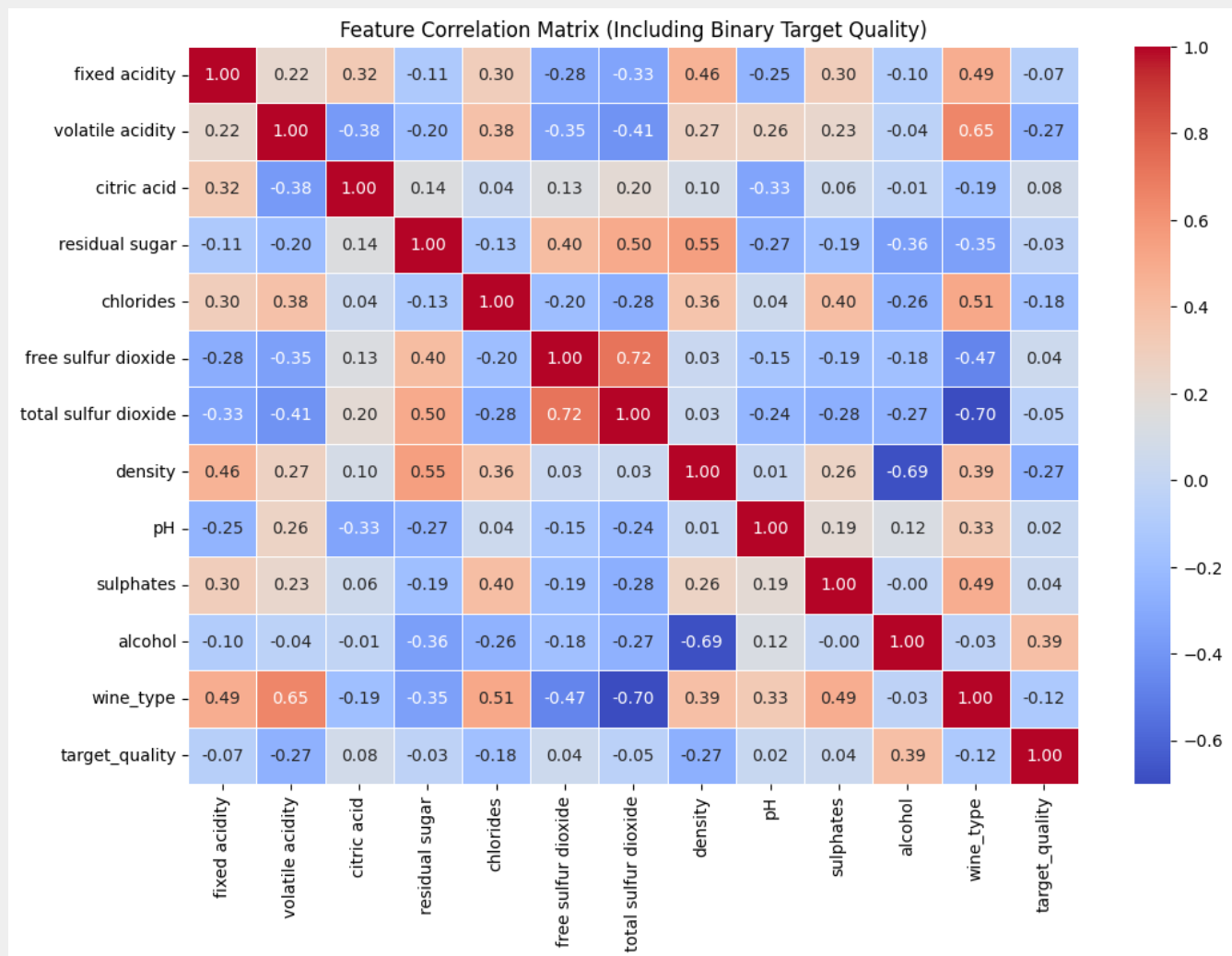
The final objective is to benchmark these geometric models against **XGBoost** to determine the most effective paradigm for predicting wine quality.

2. Dataset Preparation & Exploratory Analysis

The analysis utilizes the official Wine Quality dataset from the UCI Machine Learning Repository, combining both red and white wine subsets. The following systematic preprocessing pipeline was executed:

1. **Horizontal Integration:** The red and white wine dataframes were vertically concatenated. A binary feature `wine_type` (Red = 1, White = 0) was introduced to retain source characteristics.
2. **Target Binarization:** The original multi-class quality score (3 to 9) was converted into a binary target (`target_quality`). Samples scoring ≥ 6 are classified as **1 (High Quality)**, and scores < 6 as **0 (Standard Quality)**.

3. **Statistical Distribution:** The dataset contains a total of 6,497 instances, exhibiting a balanced distribution between high-quality and standard-quality samples.
4. **Data Splitting:** The unified dataset was partitioned into **80% training** and **20% testing** using stratified splitting (stratify=y) to maintain identical class ratios across both sets.
5. **Feature Standardization:** Since SVM relies on computing geometric distances to establish the separating hyperplane, StandardScaler was applied to normalize features. The scaler was fitted exclusively on the training partition to eliminate data leakage.



3. Classifiers' Description

In this project, three classification configurations are studied based on the theoretical principles covered in the course lectures:

- **1. Hard Margin SVM:** This classifier focuses strictly on boundary samples known as **Support Vectors** to find an optimal linear decision boundary with the maximum margin between classes. It operates under the strict assumption that the data is perfectly separable. In this implementation, the penalty parameter C is set to a very high value ($C = 10^5$) to simulate an error-free boundary constraint. To prevent the optimization solver from freezing due to real-world data overlap, a maximum iteration guard was applied, and evaluation utilizes the functional distance from the decision hyperplane.
- **2. Soft Margin SVM:** Since real-world datasets are usually overlapping and not perfectly separable, this configuration introduces flexibility by utilizing **Slack Variables**. By setting a balanced penalty parameter ($C = 1.0$), the model allows a controlled degree of error or boundary violations during training. This optimization balances maximizing the margin size while minimizing training classification errors, which prevents overfitting and improves generalization on unseen test data.
- **3. XGBoost Classifier:** Operating under a completely different paradigm than SVM, XGBoost is an advanced ensemble method based on sequential **Decision Trees**. Instead of defining a single global geometric distance boundary, it builds multiple weak trees consecutively, where each tree is trained to correct the errors made by previous ones. This enables the model to effectively capture complex, non-linear relationships among the physicochemical properties without complex boundary calculations.

4. Results and Discussion

After training the models on the normalized training subset, performance metrics were computed using the independent test set. The experimental results are summarized in the evaluation table below:

Classifier Model	Test Accuracy	F1-Score	ROC-AUC
Hard Margin SVM	0.6738	0.7430	0.7201
Soft Margin SVM	0.7346	0.8009	0.7996
XGBoost Classifier	0.7900	0.8393	0.8673

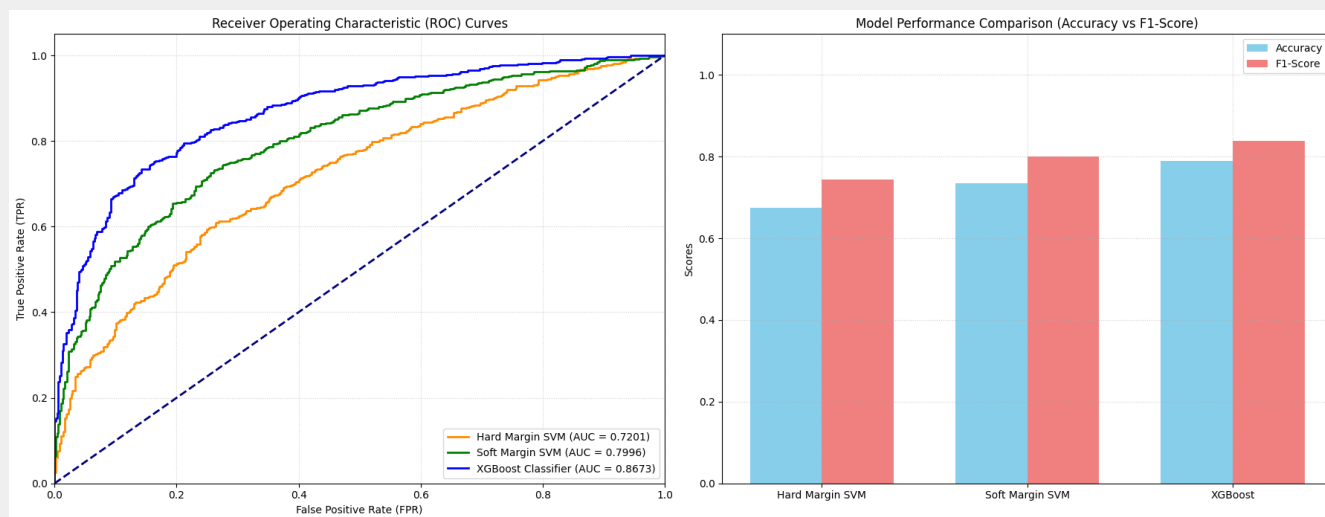
Discussion of Results

□ The **Hard Margin SVM** yielded the lowest overall performance (67.38% Accuracy and 0.7201 ROC-AUC). Because the physicochemical wine features overlap heavily, forcing a strict penalty ($C = 10^5$) with a maximum iteration limit restricted the model from full geometric convergence. However, by properly evaluating the boundary using exact decision hyperplane distances rather than unstable probability scores, it achieves a mathematically realistic ROC-AUC of 0.7201.

□ **Soft Margin Improvement:** On the other hand, the **Soft Margin SVM** ($C = 1.0$) introduced slack variables to tolerate minor boundary violations. This flexibility significantly enhanced the classifier's adaptability, increasing the Accuracy to 73.46% and the ROC-AUC to 0.7996. This clearly demonstrates that a relaxed, regularized margin generalizes much better on noisy, real-world data.

□ **XGBoost vs. SVM:** **XGBoost** easily outperformed both SVM models, achieving the highest metrics (79.00% Accuracy and 0.8673 ROC-AUC). While linear SVM models are strictly limited to drawing a single straight hyperplane to split the classes, XGBoost

builds sequential decision trees that can natively isolate and map the highly non-linear chemical relationships within the dataset.



5. Conclusion

This project successfully developed a robust pattern recognition pipeline for wine quality classification. The experimental comparison confirms that while feature standardization and margin relaxation (**Soft Margin**) significantly improve the performance and stability of distance-based geometric classifiers like SVM, these models remain fundamentally limited by linear boundary assumptions when handling highly overlapping datasets.

XGBoost proved to be the superior model for this classification task, offering the highest accuracy, F1-score, and discriminatory power (ROC-AUC). For future practical deployments, this tree-based ensemble approach is strongly recommended for physicochemical data profiling due to its native capacity to model complex, non-linear structural attributes without requiring complex kernel optimization.